Chemistry Exercises SW1 – Introduction to Substances and Elements

1) How does a chemical reaction differ from a physical process?

2) a) What do you call a heterogeneous mixture of solid and liquid?

Suspension

b) Provide an example.

c) What separation method could be applied to isolate the pure substances?

3) Which statement is false?

a) The state of matter of a substance always depends on temperature and ambient pressure.

b) A substance sublimates while releasing energy.

c) The density of a substance depends on temperature and pressure.

d) The density of a substance usually decreases with increasing temperature.

4) a) How much do 2 mol of water (H_2O) weigh in grams?

b) How many water molecules are contained in 2 mol of water?

5) State the molar mass of sodium chloride.

Na = 22.99 g/mol, Cl = 35.45 g/mol

6) How many grams of sodium chloride are needed to prepare 200 ml of a sodium chloride solution with concentration c = 500 mmol/L?

7) How many grams of sodium chloride are in 700 ml of a solution with concentration c = 1.5 mol/L?

8) You need to enrich 300 ml of culture medium with 0.5% sucrose. How many grams of sucrose are needed?

9) You want to make Limoncello at home. You have heard that to do this, you should soak lemon peel in vodka and after one week dilute it with sugar water. You are using vodka with an alcohol content (EtOH) of 40%. The Limoncello should later have an alcohol content of 25%, and you want to obtain 1 liter.

How much vodka do you need?
How much sugar water must you add after one week?

Answers: Chemistry Exercises SW1 – Introduction to Substances and Elements

1) How does a chemical reaction differ from a physical process?

In a chemical reaction, a substance changes and new substances are formed.

2) a) What do you call a heterogeneous mixture of solid and liquid?

Suspension

b) Provide an example.

Wheat beer, blood, mud

c) What separation method could be applied to isolate pure substances?

Separation through filtration, or by sedimentation / centrifugation, then decanting.

3) Which statement is false?

a) The state of matter of a substance always depends on temperature and ambient pressure. → True

b) A substance sublimates while releasing energy. → False

c) The density of a substance depends on temperature and pressure. → True

d) The density of a substance usually decreases with increasing temperature. → True

4) a) How much do 2 mol of water (H_2O) weigh in grams?

The molar mass is calculated from the atomic masses:

O = 15.999 g/mol, H = 1.008 g/mol

Water = 18.015 g/mol

2 mol = 36.03 g

b) How many water molecules are contained in 2 mol of water?

2 mol = 1.2044×10^{24} molecules

5) State the molar mass of sodium chloride.

Na = 22.99 g/mol, Cl = 35.45 g/mol → Total = 58.44 g/mol

6) How many grams of sodium chloride are needed to prepare 200 ml of a sodium chloride solution with concentration c = 500 mmol/L?

29.22 g/L × 0.2 L = 5.844 g

7) How many grams of sodium chloride are in 700 ml of a solution with concentration c = 1.5 mol/L?

87.66 g/L × 0.7 L = 61.36 g

8) You need to enrich 300 ml of culture medium with 0.5% sucrose. How many grams of sucrose are needed?

0.5 g/100 ml → 1.5 g for 300 ml

9) You want to make Limoncello at home. You have heard that to do this, you should soak lemon peel in vodka and after one week dilute it with sugar water. You are using vodka with an alcohol content (EtOH) of 40%. The Limoncello should later have an alcohol content of 25%, and you want to obtain 1 liter.

How much vodka do you need?
How much sugar water must you add after one week?

Concentration information (volume percent):
c1 (Vodka) = 40 ml/100 ml
c2 (Limoncello) = 25 ml/100 ml
V1 = ?, V2 = 1 liter → 1000 ml → calculate everything in ml!

c1 * V1 = c2 * V2 → V1 = (0.25 * 1000) / 0.4 = 625 ml → We need 625 ml of vodka.

How much sugar water? : 1 liter – 625 ml = 375 ml